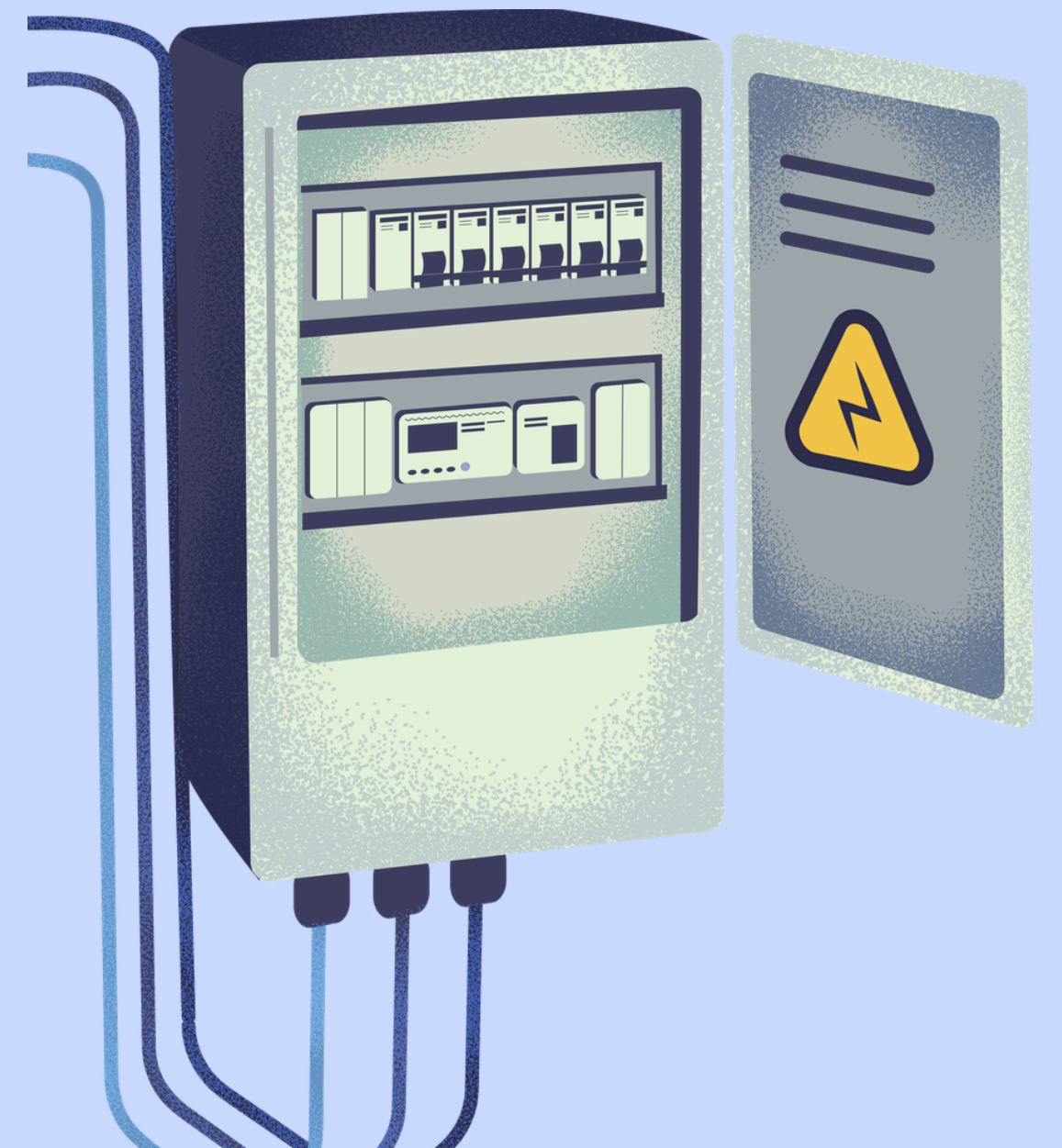


Smart electricity Anomaly Detection



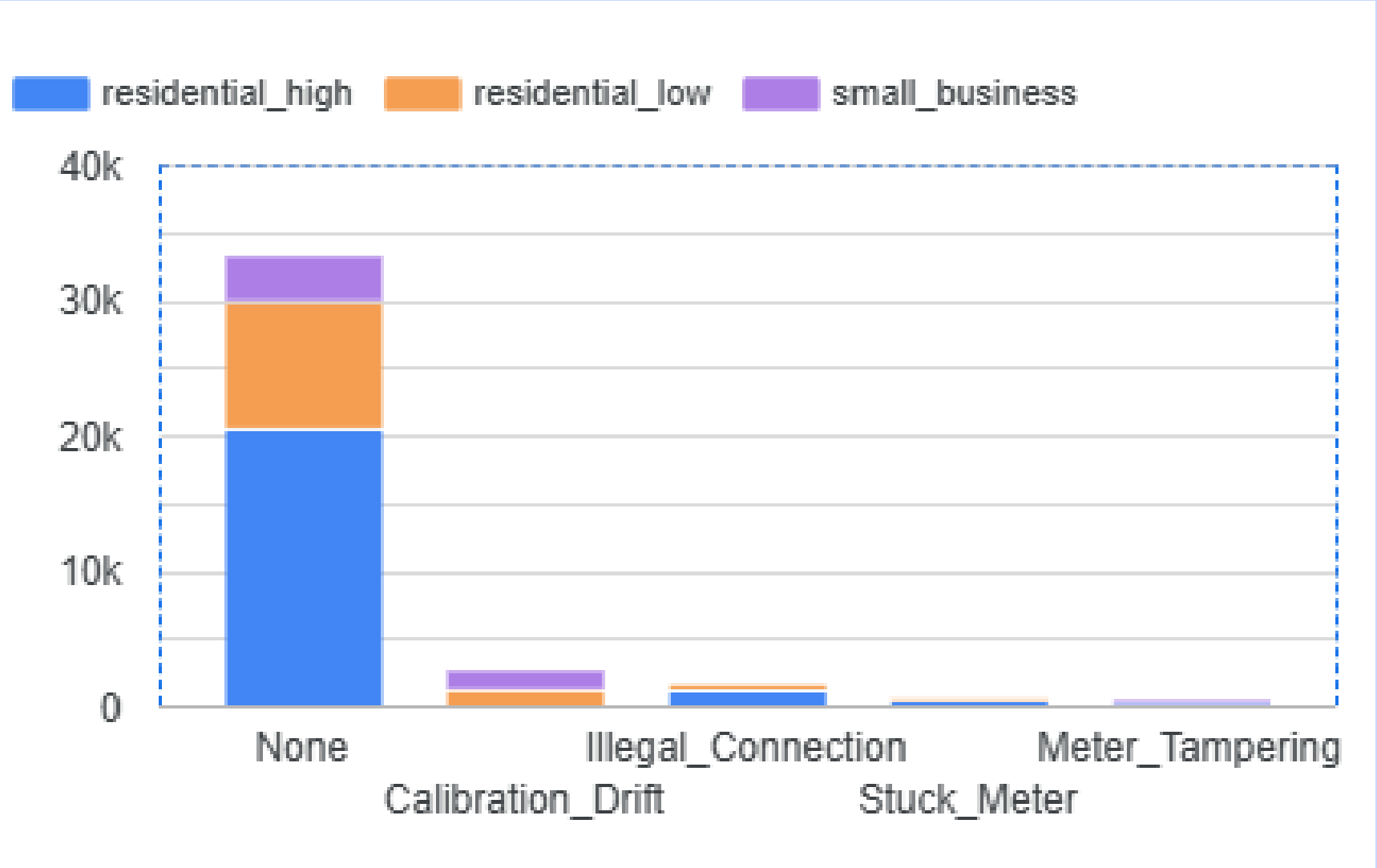
Smart Electricity Meter Anomaly Detection Overview

- **Core Metrics Callout Boxes (Top KPI Cards):**
 - 12 Unique Meter IDs Monitored
 - 22,245 Total Flagged Readings for Audit
 - 3 Primary Segments: Residential High, Residential Low, Small Business
- **Key Takeaway:**
 - Advanced machine learning anomaly detection identified tens of thousands of suspect readings across standard operating periods.
 - Auditing efforts can now be prioritized targeting high-volume accounts and specific operational anomalies

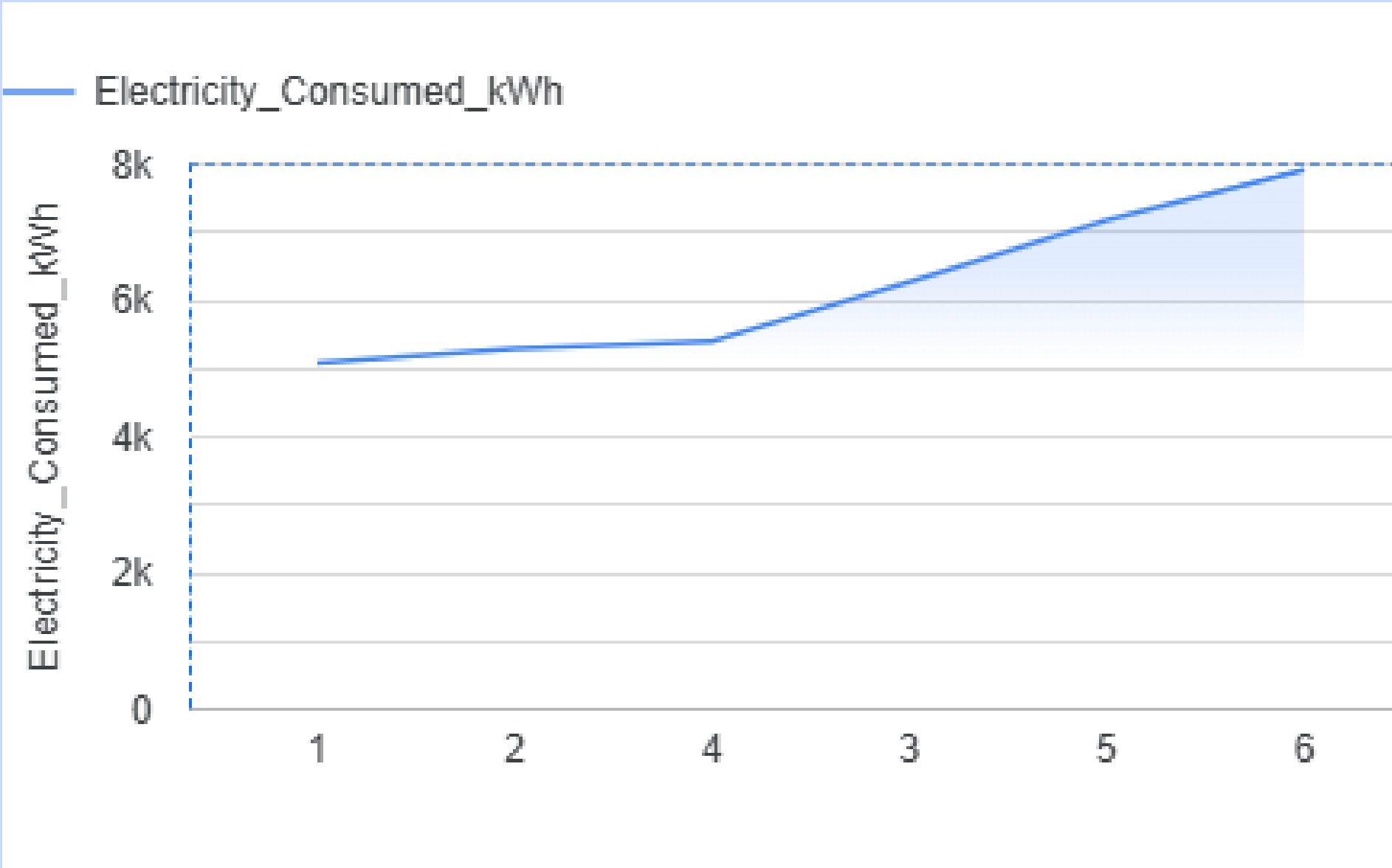
Electricity Consumption Patterns Across Segments

- **Key Insights:**
 - **Overall Trend:** Monthly electricity consumption (kWh) demonstrates steady growth from months 1 through 6, spiking in the latter half.
 - **Segment Dominance:** Residential High accounts for the vast majority of overall consumption, followed by Residential Low and Small Business.
 - **Normal Baseline:** The vast majority of total volume represents standard, unflagged consumption ("None" anomaly category).
- **Business Value:** Establishes normal consumption profiles per customer class, ensuring legitimate high-volume usage isn't incorrectly flagged as fraud.

Electricity Consumed (kWh) by Anomaly Type and Segment



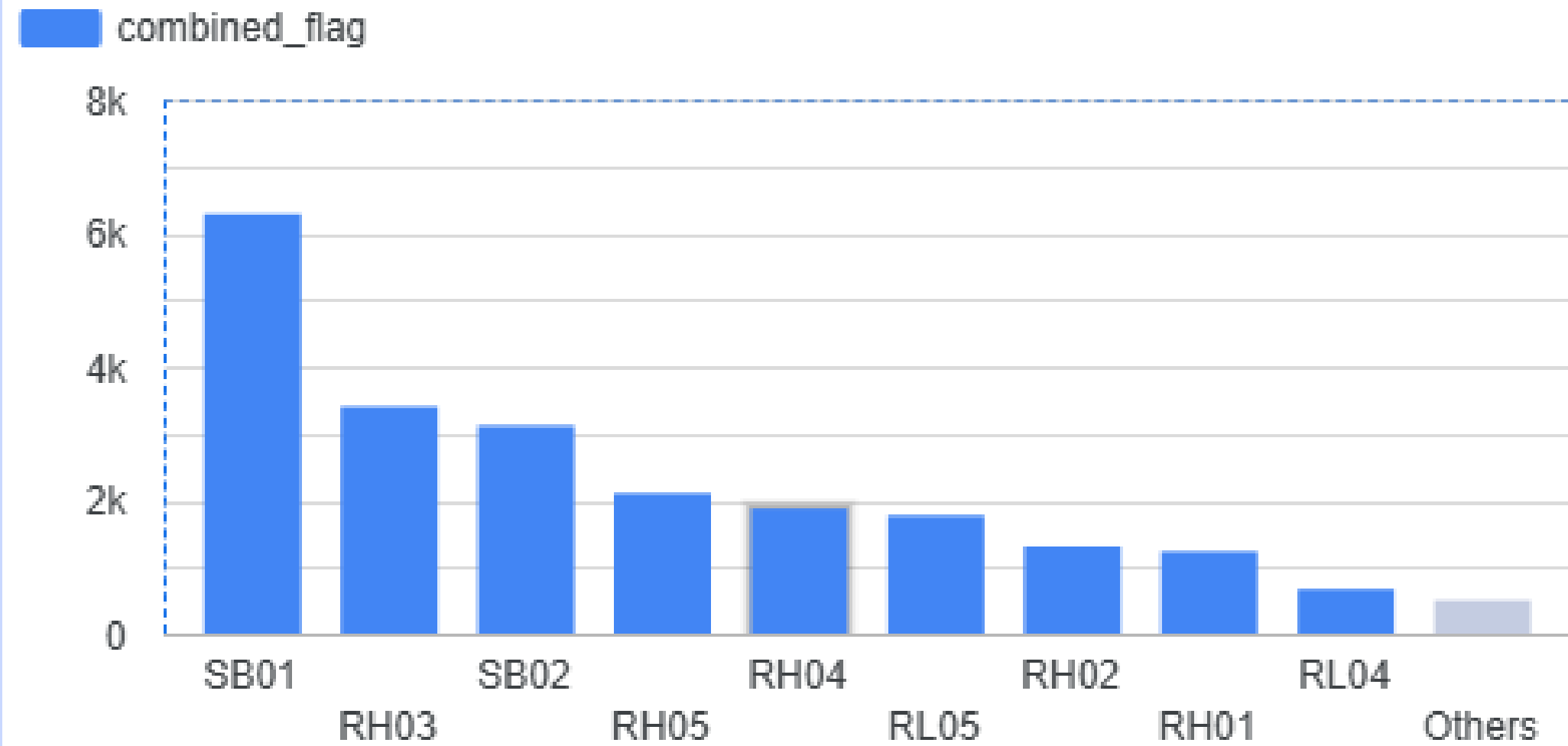
Electricity Consumed (kWh) by month



Top Accounts Requiring Immediate Field Inspection

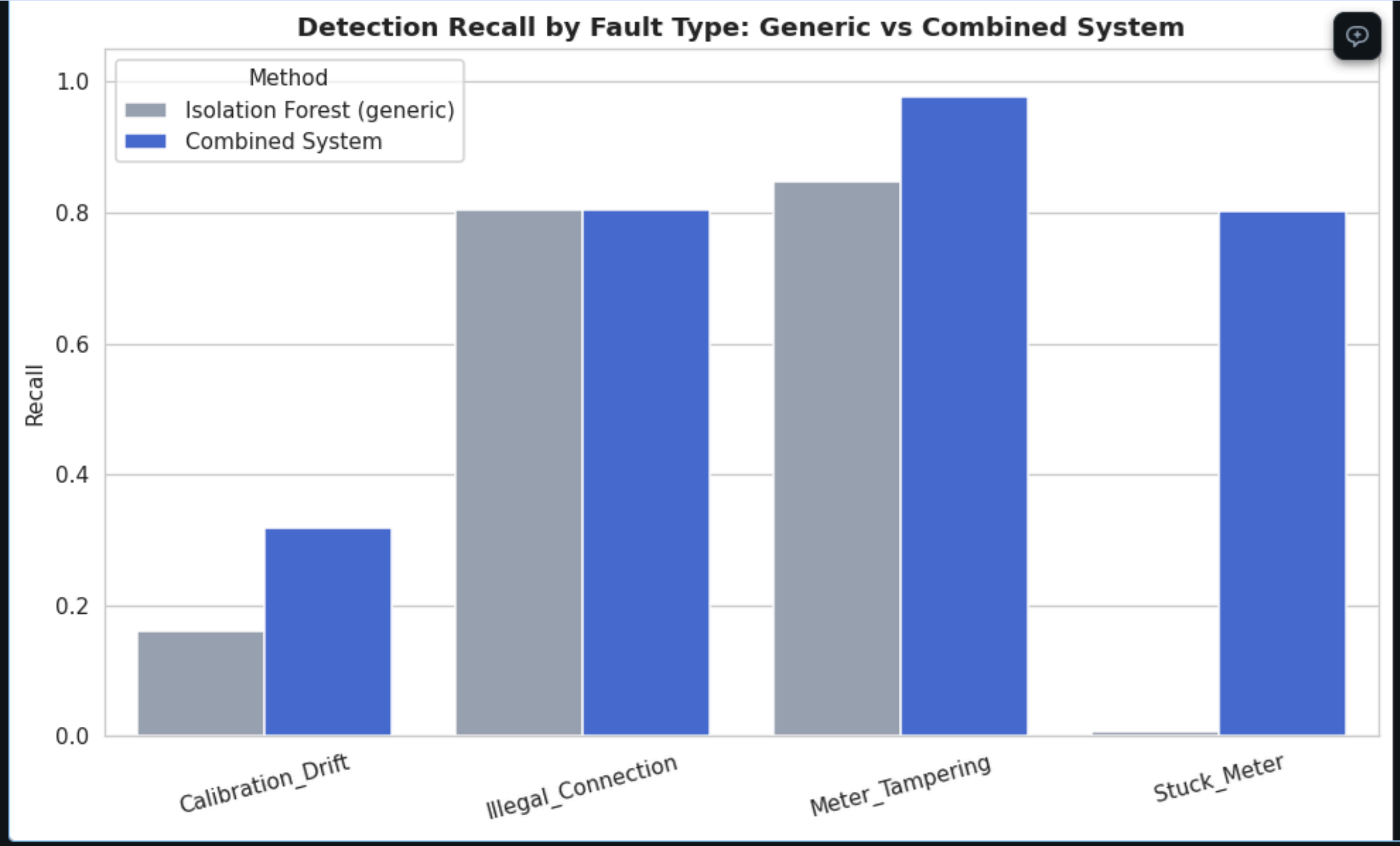
- **Key Insights:**
 - **Concentrated Risk:** Flags are heavily concentrated in a small group of high-risk meters.
 - **Top Flagged Accounts:**
 - **SB01 (Small Business 01):** Highest volume of flagged events (>6,000 flags).
 - **RH03 & SB02:** Next highest priority accounts for immediate physical audit (~3,000–3,500 flags each).
 - **RH05, RH04, RL05, RH02, RH01, RL04:** Secondary tier requiring automated or scheduled follow-ups.
- **Action Item:** Focus physical inspection teams on SB01 and RH03 first to maximize early ROI and prevent high-value revenue leakage.

Accounts Flagged for Audit



Model Performance Insights

- **Stuck Meter Breakthrough:** The Combined System fixes a critical blind spot, going from 0% to 80% detection on frozen meters.
- **Near-Perfect Fraud Detection:** Boosts Meter Tampering detection to 98% (up from 85%), ensuring high-value theft is caught.
- **Doubled Hardware Fault Capture:** Twice as effective at detecting Calibration Drift (32% vs 16%) to catch failing meters early.
- **Bottom Line:** The combined model outperforms generic models across every category without sacrificing baseline accuracy.



Overall Insights

- **Superior Fraud Capture:** The Combined System catches 98% of Meter Tampering cases, directly protecting high-value revenue from theft.
- **Eliminates Critical Blind Spots:** Successfully detects 80% of Stuck Meters—an issue generic models missed entirely (0% detection).
- **Doubles Maintenance Insights:** Reaches double the accuracy (32%) on Calibration Drift, allowing teams to fix failing hardware before it breaks.
- **Actionable Field Prioritization:** Enables dispatching audit teams directly to top-risk accounts (SB01 and RH03) with high confidence.
- **Bottom Line:** Upgrading to the Combined System directly reduces financial loss and automates field team triage.